A Centralized System for Real-Time Farm Animal Disease Reporting

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Project Proposal Report

Sandaru G A – IT22258908

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Sanadru G A	IT22258908	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

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ABSTRACT

The livestock sector plays an important role in ensuring food security and rural livelihoods in Sri Lanka. However, the current livestock disease surveillance system largely depends on manual reporting methods and reactive response mechanisms. Field veterinary officers often face challenges such as delayed disease reporting, lack of spatially precise outbreak information, and limited tools for monitoring the spread of infectious livestock diseases. As a result, disease outbreaks are frequently detected only after they have already spread across multiple locations, causing significant economic losses and difficulties in containment.

To address this issue, this research proposes a **Real-Time Geospatial Clustering and Trajectory Mapping component** within the **Animal Disease Reporting System (ADRS)**. The proposed system integrates spatial data analysis, environmental information, and predictive modelling to support early detection and monitoring of livestock disease outbreaks. The system applies **spatio-temporal clustering algorithms** to identify emerging outbreak clusters and utilizes environmental factors to estimate potential disease spread patterns. In addition, predictive models are used to forecast possible outbreak trajectories, while **Kernel Density Estimation (KDE)** is used to generate spatial risk zones across geographic regions.

The results of the analysis are visualized through an interactive geospatial intelligence map that allows veterinary authorities to monitor disease outbreaks, identify high-risk zones, and make timely intervention decisions. By transforming traditional reporting systems into a real-time spatial surveillance platform, the proposed system aims to enhance outbreak detection, improve veterinary response planning, and support more effective livestock disease management.

Keywords: Livestock Disease Surveillance, Spatial Epidemiology, Geospatial Analysis, Outbreak Prediction, Veterinary Decision Support

Table Contents

DECLARATION i
ACKNOWLEDGEMENT ii
ABSTRACT iii
LIST OF FIGURES v
LIST OF ABBREVIATIONS v
1. INTRODUCTION 1
1.1 Background and Context 1
1.2 Literature Review 3
1.3 Research Gap 5
2. OBJECTIVE 7
2.1 Main Objective 7
2.2 Specific Objectives 7
3. METHODOLOGY 8
3.1 Research Approach 8
3.2 Technical Methods and Algorithms 8
3.3 Tools and Platforms 9
3.4 Data Collection and Ethical Considerations 10
3.5 Validation Strategy and Performance Metrics 11
4. HIGH-LEVEL SYSTEM ARCHITECTURE 12
4.1 Overview of Proposed Solution and its Components 12
4.3 Component Interactions 14
4.4 Data Flow Explanation 15
4.5 Scalability and Security Considerations 16
5. IMPLEMENTATION 17
6. USER REQUIREMENTS 19
6.1 Stakeholder Identification 19
6.2 Functional Requirements 19
6.3 Non-Functional Requirements 20

7. COMMERCIALIZATION PLAN 22
8. BUDGET AND JUSTIFICATION 27
9. WORK BREAKDOWN STRUCTURE (WBS) 28
10. GRANTT CHART 32
11. REFERENCES 33

LIST OF FIGURES

LIST OF ABBREVIATIONS

Abbreviation	Definition
ADRS	Animal Disease Reporting System
API	Application Programming Interface
AUPRC	Area Under Precision-Recall Curve
B2B	Business-to-Business
B2G	Business-to-Government
BCS	Body Condition Scoring / Body Condition Scores
BSS	Bovine Severity Score
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNNs	Convolutional Neural Networks
CoT	Chain of Thought
CV	Computer Vision
DAPH	Department of Animal Production and Health
DSR	Design Science Research
FMD	Foot-and-Mouth Disease
IoU	Intersection over Union
LSD	Lumpy Skin Disease
MAE	Mean Absolute Error
MLP	Multi-layer perceptron
OOD	Out-of-Distribution

1. INTRODUCTION

1.1 Background and Context

Food economy in the world is center-stage with livestock production systems. In both the developed and developing world, livestock play a significant role in food security, agricultural jobs, and livelihoods of the rural populations. Animal products, especially milk, meat and eggs, continue to play a significant role as rich sources of quality protein and micronutrients to billions of people. With the ever-increasing global populations and urbanization, there has been a consistent rise in the demand of animal protein throughout the last few decades, which has burdened the livestock production systems in order to remain productive and resilient. According to the estimates made by the Food and Agriculture Organization, the livestock can help provide the livelihoods of over 1.3 billion people all over the world and contribute about 40 percent of the worldwide agriculture GDP [1]. The livestock have wider social roles in rural societies that can be described as having economic value, in that livestock are financial resources, sources of household income, and economic insurance. However, as this dependence on livestock systems increases, agricultural economies are also highly vulnerable to threats of disease outbreaks.

In South Asia, livestock production assumes a relatively significant role in the rural livelihoods and local food systems. Although the livestock industry of Sri Lanka is relatively small compared to its crop agriculture counterpart, it has steadily grown over the recent years as part of the national initiative to bolster the local dairy production and decrease the dependency on imported milk derivatives. National agricultural statistics indicate that livestock subsector is a contributor to national GDP of Sri Lanka of about 1.2-1.5% and supports the livelihood of hundreds of thousands of smallholder farmers involved in dairy and livestock farming [2]. Dairy farming has especially been cited as one of the priorities in agricultural development, with government efforts supporting production of domestic milk because of the rising consumer demand. A large portion of the sector is typified by small holder in production systems with farmers keeping a small number of cattle or buffalo through mixed crop-livestock production systems. Although such decentralized production systems offer flexibility and income-generating opportunities to rural families, this results in difficulty in disease control and veterinary control because livestock populations are geographically spread to thousands of small farms.

One of the biggest menaces to livestock production systems that have shown to be harder to handle are infectious diseases with the ability to propagate among animals as well as geographic areas rapidly. The economic costs in the livestock producers are still high due to diseases like Foot-and-Mouth Disease (FMD), Lumpy Skin Disease (LSD), and Subclinical Mastitis (SCM). One such example is FMD, which is commonly known as one of the most transmissible viral infections affecting the cloven fed animals and can spread rapidly by direct contact, airborne, or contaminated materials [3]. Outbreaks may cause serious losses in milk production, limitation of animal flow, and a massive loss in the economy. Lumpy Skin Disease has also become one of the

greatest threat to the cattle populations in some Asian nations, which often leads to skin wounds, low milk production, infertility and occasionally fatalities to infected cows [4]. Even the conditions that might not be noticeable at a specific moment like subclinical mastitis can seriously stringent the milk productivity and profitability in a farm. On the farm level, the diseases have a direct impact on household income and food production, and on the national level, mass outbreaks can destroy livestock supply chains and derail strategies of agricultural development.

Even though the magnitude of these risks is high, livestock disease surveillance systems in most parts have been largely in the form of a response than proactive. Practically, the process of disease reporting is usually based on the late communication between farmers, veterinary officers, and the administrative bodies. Field reports could be sent by paper-based reports, phone calls or regular visits to the veterinary thus creating disjointed and time-delayed data collections. Lack of a single surveillance hub means that the information on the disease can be scattered across various institutions and authorities are not able to come up with a complete overview of the outbreak trends. Lack of real-time monitoring systems means that veterinary services are also unable to detect spatial clusters of infection as well as respond quickly to any possible outbreak. To the smallholder farmers, such delays may mean massive losses to them in terms of the spread of diseases before the efforts to control them are realized.

Recent developments in digital agriculture and geospatial technologies can provide a promising solution to these problems. Over the past few years, geographic Information Systems (GIS), spatial data analytics, and interactive mapping systems have been embraced in epidemiological studies to investigate the way diseases disseminate in landscapes and populations [5]. GIS offers the benefit of combining spatial data with disease surveillance systems to researchers and policymakers to locate outbreak hotspots, visualize gender patterns, and assess environmental conditions that promote the development of diseases. The most effective interface that enables the conversion of intricate epidemiological data into intuitive maps and spatial dashboards is interactive geospatial visualization tools, especially. With these systems, users have the ability to track the occurrence of diseases geographically as well as analyze the emerging trends in a manner that will be instantly available to decision-makers.

Based on these technological advances, the current paper suggests a Centralized System of Real-Time Farm Animal Disease Reporting that will enable fast disease surveillance and veterinary response. The main element of this system is Real-Time Geospatial Visualization by means of Interactive Intelligence Map that translates the incoming disease reports into the dynamic spatial representation of the livestock health condition. Instead of keeping the reports of the diseases in the role of mere written records, the system allows the veterinary authorities to see the reported cases on a map as they happen, detecting a new outbreak of infection, and estimating the possible risk areas in real-time. Through the transformation of disjointed farm-level observations into the constantly updated spatial intelligence, the proposed system will enhance the disease surveillance

functions and contribute to making more efficient decisions in the framework of livestock health management systems.

1.2 Literature Review

Surveillance of livestock diseases is a growing area of research in veterinary epidemiology and agricultural informatics as well as digital agriculture. Foot-and-Mouth Disease (FMD), Lumpy Skin Disease (LSD) and other transboundary animal diseases are infectious livestock diseases that pose a great threat to livestock productivity, rural livelihood, and the food security system of a country. The threat of disease outbreak also rises as the production of livestock grows all over the world as a result of the growing consumption of animal protein. Good systems of surveillance are hence very important in early detection of outbreaks, disease surveillance and quick veterinary response. The past few years have seen a surge in the adoption of digital farming systems such as mobile reporting, cloud computing based information systems, and geospatial data platforms, which have generated new possibilities to enhance livestock disease surveillance systems. Geographic Information Systems (GIS) and spatial analytics are among such technologies that have been identified as important tools to track the spatial patterns of disease outbreaks and detecting high-risk geographic areas [4].

Spatial epidemiology methods have been examined in a vast amount of research to investigate dynamics of livestock diseases. Alkhamis et al. (2019) examined the genetic sequencing of Foot-and-Mouth Disease and the spatial modelling technologies to study the phylogeographic distribution in the Middle East and South Asia [5]. Their results indicated that the spread of diseases is commonly traced on the livestock trade patterns and environmental routes instead of administrative ones. Although the study had some informative information about the regional transmission, the analysis was based on retrospective viral sequence data, thus, not useful in real-time management of the outbreak. On the same note, Laffan and Ward (2021) used spatial scan statistics to identify the cluster of bovine respiratory disease outbreaks in the Australian cattle population [6]. Their study was able to find statistically significant outbreak clusters with the use of spatial scanning. The model was, however, based on periodic aggregated data, as opposed to real-time: continuous reporting facilities, which makes the model less useful in detecting outbreaks early.

Recent studies have delved into machine learning solutions to enhance disease monitoring and prediction. The article by Tran et al. (2022) presents a gradient-boosted machine learning model trained to forecast the Lumpy Skin Disease risk location in southeast Asia with regard to

environmental characteristics, including land cover, cattle density, and climatic conditions [7]. The model showed a great predictive power and proved the significance of environmental variables in the distribution of the disease. The outputs were however not dynamic monitoring tools that can react to new disease reports but rather were static predictive maps which were updated periodically. On the same note, Punyapornwithaya et al. (2023) used ecological niche modelling to examine 669 LSD outbreaks in Thailand in the MaxEnt modelling framework [8]. Among the key predictors of the risk of outbreaks the study has found the land cover and cattle density. Whereas the research enhanced the knowledge on drivers of environmental diseases, it was only more on the spatial modelling and not the operational disease monitoring systems.

Besides scholarly research, there are a number of international websites that have been established to monitor livestock diseases around the world. The EMPRES-i system managed by the Food and Agriculture Organization gathers animal disease information in numerous countries and maps diseases occurrence across the globe [9]. Similarly, the World Organisation for Animal Health has the World Animal Health Information System (WAHIS) which compiles official reports of diseases provided by the member countries and releases outbreak data to international monitoring [10]. These platforms offer quality infrastructures regarding reporting of diseases all over the world and also internationally aid in coordinating disease control initiatives. Their main purpose is however, data aggregation as opposed to real time surveillance. The reporting processes many times entail the process of verification and review by the administration which creates delay and restricts its use in instant detection of outbreak at the regional or farm levels.

It has also been suggested to use mobile veterinary reporting systems to enhance the disease data collection in rural setting. In a study of a community- based mobile surveillance platform in Uganda, Lindahl et al. (2019) discovered that the digital reporting system has a significant positive effect on detecting the disease compared to the earlier paper-based reporting [11]. In the same way, Hassan et al. (2021) explored the digital livestock health monitoring instruments occupied by veterinary officers and community animal health workers in Pakistan and found that there was a better connection between farmers and veterinary authorities [12]. Although these systems have greatly improved on efficiency in reporting, most of them are still constrained to data collection. Centralized databases are usually the place where disease reports are stored, which lacks the ability to perform spatial analysis or geospatial visualization.

An overview of these studies shows that there are a number of similar methodological trends. A number of researchers use GIS-based modelling and spatial clustering to analyze the distribution patterns of disease, through predictive models they are based on either statistical or machine learning algorithms to predict disease risk based on environmental factors. In the meantime, the

main priority in digital surveillance platforms is to enhance the effectiveness of reporting and data management in diseases. Whilst these approaches do add capabilities of value each in their own right, they tend to be disjointed. Digital reporting systems are rarely integrated with interactive geospatial visualization tools, and analytical spatial models are rarely combined with infrastructure of real-time reporting.

Such restrictions make the current livestock disease surveillance systems incredibly unproductive. Disjointed surveillance systems slow down the identification of outbreaks, and the absence of spatial intelligence capabilities of integrating the spatial intelligence of veterinary authorities. In the absence of interactive geospatial visualization tools, the decision-makers might not be able to interpret the outbreak patterns fast enough to adopt the timely containment measures.

The gap in research is therefore evident in the literature. The literature is replete with literature that has concentrated on either disease reporting systems or spatial epidemiological modelling but little has been done combining both real-time reporting and dynamic geospatial visualization. In this regard, it is suggested that the current study will fill this gap by proposing a Centralized System, Real-Time Farm Animal Disease Reporting with Real-Time Geospatial Visualization using an Interactive Intelligence Map. The proposed system will change the scattered disease reports into operational geospatial intelligence by combining the centralized reporting infrastructure with interactive spatial analytics to assist in the process of monitoring outbreaks, spatial analysis, and veterinary decision-making.

1.3 Research Gap

Research Gaps

Several critical limitations exist in current livestock disease surveillance research and operational systems. Although recent studies have applied spatial epidemiology and GIS-based analysis to understand disease distribution, many of these approaches are implemented as isolated analytical models rather than integrated surveillance platforms. Key research gaps include:

- **Absence of real-time integrated surveillance systems:** Most existing systems record disease reports retrospectively and do not provide real-time monitoring capabilities.
- **Low spatial resolution of outbreak data:** Disease information is commonly aggregated at district or regional levels, resulting in significant geographic uncertainty during outbreak detection.

- **Lack of predictive outbreak analysis:** Current studies mainly analyze historical outbreak data and rarely include models that forecast disease spread trajectories.
- **Limited operational geospatial visualization:** Many surveillance platforms store disease records but lack interactive spatial dashboards that support decision-making by veterinary authorities.

Why Current Approaches Are Insufficient

The existing systems of livestock disease surveillance are still inadequate because of a number of operational and structural factors. These constraints decrease their abilities to detect the outbreak early and assist veterinary decision-making.

- **Slow disease reporting systems-**The global disease reporting systems (WAHIS), like the World Animal Health Information System, and the EMPRES-i in the FAO are based on administrative reporting systems that create delays between the identification of an outbreak and the publication of data [15], [16].
- **Disjointed surveillance infrastructure-**Multiple institutions tend to collect and store disease information using different databases. Such fragmentation inhibits effective data transfers and diminishes the capacity to come up with centralized outbreak intelligence [13].
- **Poor real-time outbreak surveillance-**Majority of spatial epidemiology studies are dependent on the past outbreak records and batch processing approaches. These techniques are applicable to the retrospective analysis but fail to serve to monitor live outbreaks [10].
- **Absence of spatial decision-support tools-**Most digital livestock health reporting systems are mainly concerned with enhancing data acquisition but lack the ability of spatial decision support [17].
- **Lack of support to respond to veterinary quickly-**The absence of built-in spatial visualization capabilities could mean that the veterinary authorities will be unable to recognize outbreak clusters fast, assess geographic risk zones, and take appropriate containment actions in time [14].

Consequently, the current surveillance methods tend to be unable to deliver the operational intelligence needed to manage diseases outbreaks.

Novelty and Creativity of the Proposed Study

The proposed study introduces an integrated geospatial surveillance framework that combines real-time disease reporting, spatio-temporal clustering, environmental modelling, and predictive outbreak visualization. By applying spatial clustering algorithms to detect outbreak clusters and integrating environmental variables to model disease spread patterns, the system provides a more

comprehensive understanding of outbreak dynamics. In addition, the development of an interactive geospatial intelligence map enables veterinary authorities to visualize disease clusters, predicted spread trajectories, and risk zones in real time. This integrated approach transforms traditional retrospective surveillance into a proactive decision-support system for livestock disease management

Evidence-Based Case to the Proposed System.

Previous studies in spatial epidemiology and GIS-based disease surveillance have demonstrated that spatial clustering techniques and geospatial analysis can effectively identify disease hotspots and transmission patterns. However, most existing research focuses primarily on retrospective analysis rather than operational surveillance systems. Furthermore, studies on livestock diseases such as Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD) highlight the importance of environmental and spatial factors in disease transmission. These findings indicate the need for integrated surveillance platforms that combine spatial analysis, environmental modelling, and real-time visualization. Therefore, the proposed system addresses a critical gap by developing a comprehensive geospatial surveillance framework designed to improve livestock disease monitoring and outbreak response.

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2. OBJECTIVE

2.1 Main Objective

The goal of this study is to design and evaluate a real-time geo-spatial visualisation component that turns a livestock disease report into an interactive outbreak intelligence map to support timely veterinary surveillance and decision-making.

2.2 Specific Objectives

- To create a centralized pipeline that will collate and process real-time reports of livestock diseases from field veterinary officers.

- To implement a spatio-temporal clustering model that can automatically identify and detect clusters of active livestock disease outbreaks from incoming reports. An interactive geospatial visualization module that constantly maps the incidence and spatial outbreaks of diseases in the administrative region.

- To integrate environmental and spatial data sources to analyse potential disease spread directions and generate predictive outbreak trajectories.

- To generate a continuous geospatial risk surface using spatial density estimation methods to identify high-risk livestock disease areas.

- To evaluate the system's performance and decision-support effectiveness for veterinary authorities in monitoring and responding to livestock disease outbreaks.

3. METHODOLOGY

3.1 Research Approach

This study uses a Design Science Research (DSR) approach to design, develop and evaluate a component for Real-time Geospatial Visualisation for a Centralised Livestock Disease Surveillance System called the Interactive Intelligence Map. The Design Science Research method is appropriate because the research problem requires building a technical artefact to solve a real-world problem, livestock disease monitoring. Instead of just studying what already exists, the new system is developed which will process disease reports and convert them into actionable geospatial intelligence.

The research takes the Design Science Research framework, proposed by Hevner et al. in 2004, which consists of artefact creation and evaluation, along with contribution of knowledge. This project's artefact is the Internet Intelligence Map, a platform allowing for real-time disease reporting and geospatial analysis. The veterinary system is subjected to testing for technical performance, spatial prediction accuracy and usability with stakeholders. The study can thereby solve a real-world surveillance problem while also producing valuable design knowledge for digital agriculture and spatial epidemiology.

3.2 Technical Methods and Algorithms

The system merges numerous functions for disease surveillance which make it spatially temporally and graphical descriptive.

• **Spatio-Temporal DBSCAN (ST-DBSCAN) clustering**

ST-DBSCAN identifies disease outbreak clusters that are spatially and temporally close to each other. Parameters specific to the disease, including spatial radius (εs) and temporal window (εt), will permit the algorithm to detect transmission clusters of FMD and LSD.

• **Environmental Vector Field (EVF) modelling**

An Environmental Vector Field model is applied to evaluate possible directions of disease spread. This model combines environmental variables such as wind direction, terrain slope, river flow direction, livestock density gradients and vegetation indices to produce a composite spread vector at each detected outbreak cluster.

•**Ridge Regression trajectory prediction models**

For each disease, disease-specific ridge regression models are employed for predicting the movement of disease outbreaks in time and space. These models use spatial and environmental factors and time horizons to estimate future position of outbreak centroids in the East-West and North-South dimensions.

• **Kernel Density Estimation (KDE) risk surface modelling**

A slightly adjusted version of Kernel Density Estimation (KDE) creates the risk surface of disease (i.e. risk of encountering disease) in Sri Lanka in a continuous manner. The spatial representation of the risk of disease is improved with the consideration of data quality and time decay weights.

All of these algorithms together allow the system to recognize clusters of outbreaks, estimate the direction of spread, predict the movement of outbreaks in time and space, and create risk surfaces for the purpose of decision support.

3.3 Tools and Platforms

The described system incorporates a cutting-edge, web-based geospatial technology stack for real-time data streaming and interactive data technology outlined below.

• Primary programming is done in **Python**, given the language's rich ecosystem in spatial analysis and machine learning.

• **FastAPI** is utilized for back-end API and WebSocket server development, thus providing real-time interaction between the system and users.

• The spatial database for disease report storage and geospatial queries is set using **PostgreSQL with PostGIS.**

• **The machine learning and regression modeling and clustering algorithms are** implemented through the Scikit-learn library.

• For spatial data, **GeoPandas and Shapely** are used.

• For interactive web mapping, **Leaflet.js** is used because it allows for lightweight, mobile-optimized geospatial visualization for field users.

• For the purposes of providing system scalability and reproducibility, **Docker** is used for deployed containerization.

The selected technologies possess the desired attributes of real-time processing, geospatial analysis, and scalability for web-based surveillance systems..

3.4 Data Collection and Ethical Considerations

Data Collection

The system integrates multiple data sources to support livestock disease surveillance and outbreak prediction.

Data Sources

• **Livestock disease reports** – disease cases submitted by farmers or veterinary officers through the ADRS mobile reporting interface.

• **Farm location data** – geographic coordinates (latitude and longitude) obtained through map-based location selection during case reporting.

• **Animal information** – basic livestock details such as species and approximate herd size associated with the reported case.

• **Environmental data** – real-time weather variables including wind speed, wind direction, temperature, humidity, and rainfall obtained from public weather APIs such as Open-Meteo.

• **Geospatial datasets** – terrain elevation (SRTM Digital Elevation Model), river flow direction (HydroSHEDS), livestock density maps (FAO GLW4), and vegetation indices (MODIS NDVI satellite data).

• **Administrative boundary data** – GN Division and district polygon datasets used for spatial aggregation and visualization.

Data Collection Methods

• Mobile reporting through the ADRS application.
• Retrieval of environmental data through public APIs.
• Downloading and storing open geospatial datasets for spatial analysis.
• Integration of existing livestock and geographic datasets from public repositories.

Ethical Considerations

The system follows standard research ethics and data protection principles.

• **Data privacy protection** – farm location and farmer identity information are stored securely and protected through access control mechanisms.

• **Confidentiality** – exact farm coordinates are not displayed in the visualization interface; only aggregated administrative areas are shown.

• **Responsible data use** – collected data are used strictly for livestock disease surveillance and academic research purposes.

• **Data security** – all datasets are stored within a secured database environment with restricted access.

• **Compliance with regulations** – the study follows institutional research guidelines and national data protection regulations.

3.5 Validation Strategy and Performance Metrics

Validation Strategy

The system is evaluated using multiple validation approaches to assess prediction reliability and system performance.

• **Historical outbreak data testing** – past outbreak datasets are used to train and test the prediction model.

• **Holdout validation** – datasets are divided into training and testing subsets to evaluate prediction accuracy on unseen data.

• **Historical backtesting** – previous Sri Lanka outbreak sequences are used to evaluate whether the system correctly predicts the next affected region.

• **Baseline comparison** – prediction results are compared with simple baseline models to determine whether the proposed model improves prediction accuracy.

Performance Metrics

The following measurable indicators are used to evaluate system performance.

• **Prediction accuracy** – measured using Mean Absolute Error (MAE) between predicted outbreak locations and actual outbreak locations.

• **District prediction accuracy** – percentage of correctly predicted next affected districts during historical backtesting.

• **Cluster detection accuracy** – ability of the system to correctly identify outbreak clusters using spatial clustering algorithms.

• **System response time** – time required for the system to process a new disease report and update the visualization dashboard.

• **Visualization usability** – ease of interpretation of risk maps and outbreak patterns by veterinary authorities.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and its Components

The Animal Disease Reporting System (ADRS) is a centralized livestock disease surveillance platform designed to transform field-level disease reports into actionable intelligence for veterinary authorities. The system integrates four complementary research components: Seasonal Disease Forecasting, Real-Time Geospatial Visualization, Wellness and Management Standardization, and Computer Vision–based Disease Diagnosis.

Within this architecture, the Real-Time Geospatial Visualization component, referred to as the Interactive Intelligence Map, serves as the spatial intelligence layer of the system. This component processes validated disease reports, spatial analysis outputs, and predictive modelling results to generate real-time outbreak maps, cluster visualizations, and spatial risk indicators. By converting distributed disease reports into interactive geospatial insights, the module enables veterinary authorities to monitor outbreak dynamics and support timely disease control and

response decisions.

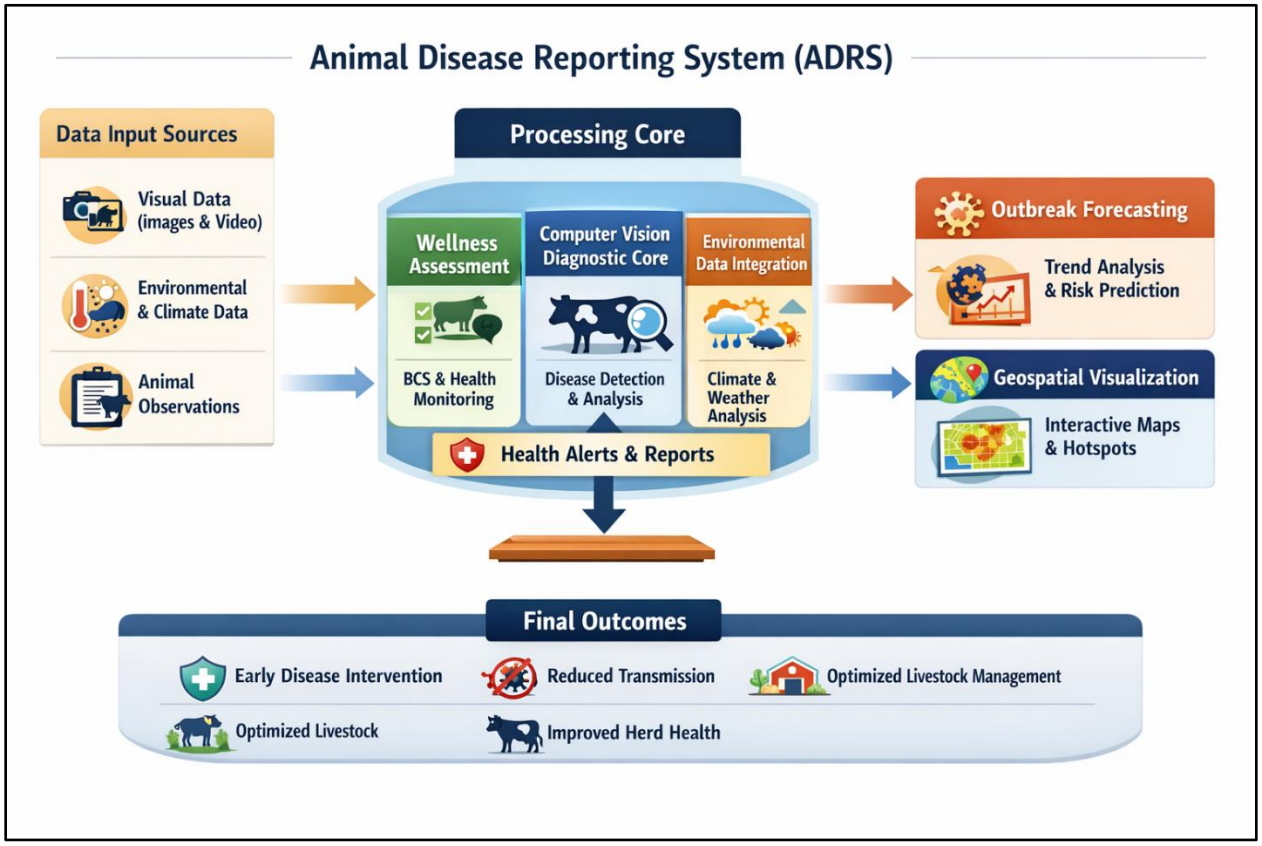


Figure 1 System Diagram with Components

4.2 Highlight of Individual Component and its Parts

The individual research component of this study is the **Real-Time Geospatial Visualization and Intelligence Module**, implemented through the PISTES analytical pipeline.

This component receives validated disease case data, geographic coordinates, clustering outputs, and predicted spread trajectories from analytical modules within the system. Using these inputs, the module generates an **interactive intelligence map** that displays current outbreak clusters, predicted spread directions, and risk zones across administrative regions.

The visualization layer transforms complex epidemiological outputs into easily interpretable map-based information, enabling veterinary officers to quickly identify high-risk areas and implement timely disease control measures.

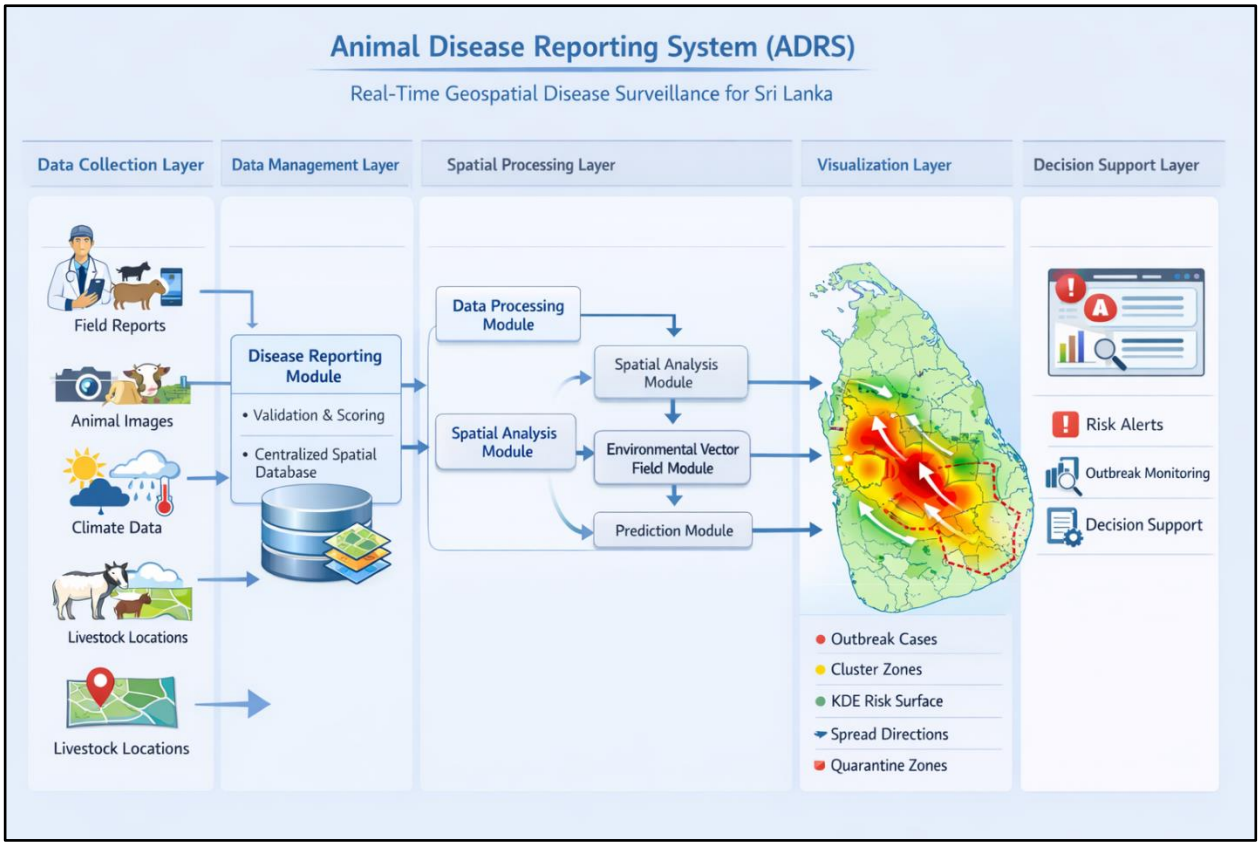


Figure 2 Pipeline of Computer Vision

4.3 Component Interactions

The ADRS components interact sequentially within the disease surveillance workflow.

1. A farmer submits a disease report through the mobile application including a farm location and animal photograph.
2. The Computer Vision module analyzes the image and generates a disease classification with a confidence score.
3. A veterinary officer reviews the report and confirms the disease case through the confirmation portal.
4. The confirmed case data is stored in the centralized PostGIS spatial database.
5. The PISTES analytical module retrieves confirmed case data and performs spatio-temporal clustering to detect outbreak clusters.
6. Environmental variables are integrated to calculate potential disease spread direction.

7. The trajectory prediction module estimates future outbreak locations for multiple time horizons.
8. Kernel density estimation generates spatial risk zones representing outbreak probability across administrative regions.
9. The visualization module renders these outputs as an interactive intelligence map.
10. Veterinary authorities access the map through the dashboard to monitor outbreaks and coordinate response actions.

4.4 Data Flow Explanation

The ADRS data flow follows a structured pipeline from data acquisition to decision support.

Input Data

• Disease reports submitted through the mobile application
• Farm location coordinates and timestamps
• Animal species and herd information
• Environmental data including weather and climate variables

Processing Stage

• Image validation and disease classification using computer vision
• Data validation and storage in the centralized spatial database
• Spatial clustering of confirmed disease cases
• Environmental modelling and outbreak trajectory prediction

Analysis Stage

• Identification of outbreak clusters
• Estimation of disease spread direction
• Generation of risk zones through kernel density estimation

Output Stage

• Interactive intelligence map displaying outbreak clusters
• Risk zone visualization across administrative boundaries
• Predicted disease spread trajectories
• Alerts and decision support information for veterinary authorities

4.5 Scalability and Security Considerations

Scalability

- Modular system architecture allows independent expansion of system components.
- The centralized database supports large-scale spatial data processing.
- Cloud-based deployment enables scaling to handle increasing disease reports.
- The system can be extended to support multiple provinces and additional livestock diseases.

Security

- Role-based access control restricts system access according to user roles.
- Secure authentication mechanisms protect system login access.
- Sensitive farm location information is protected within the database.
- Data transmission between system components uses encrypted communication protocols.
- Regular data backup and monitoring mechanisms ensure system reliability and integrity.

5. IMPLEMENTATION

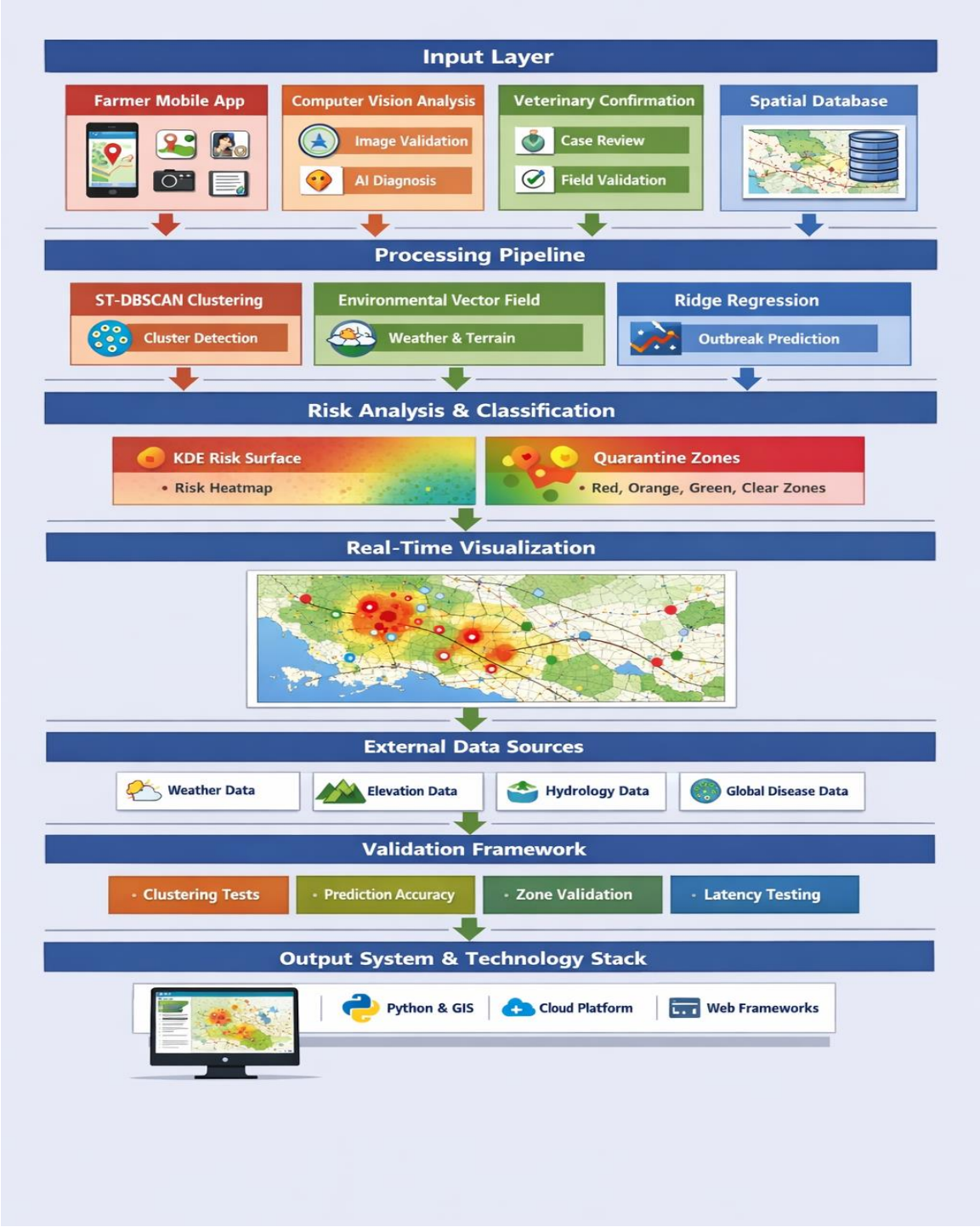


Figure 3 Geospatial Disease Analysis and Prediction Pipeline

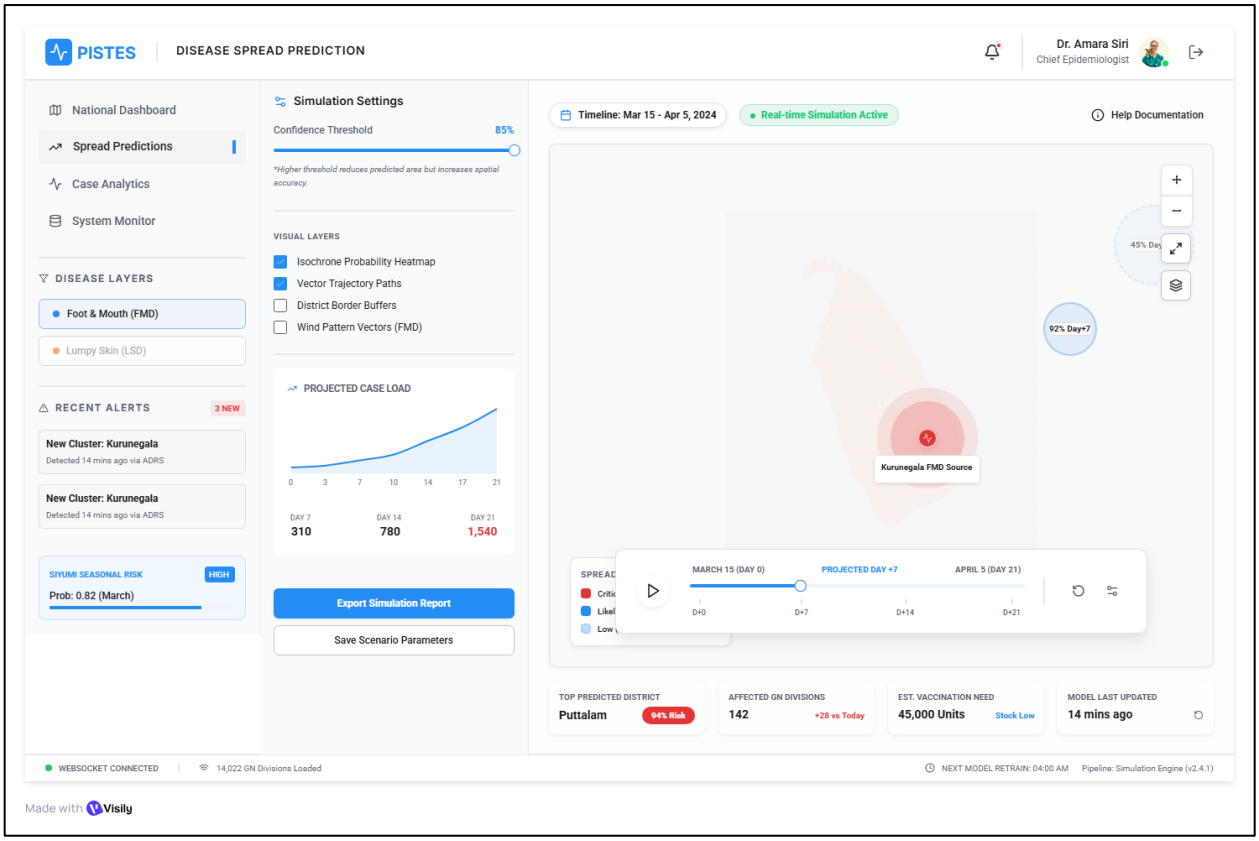


Figure 4 Trajectory View UI mockup

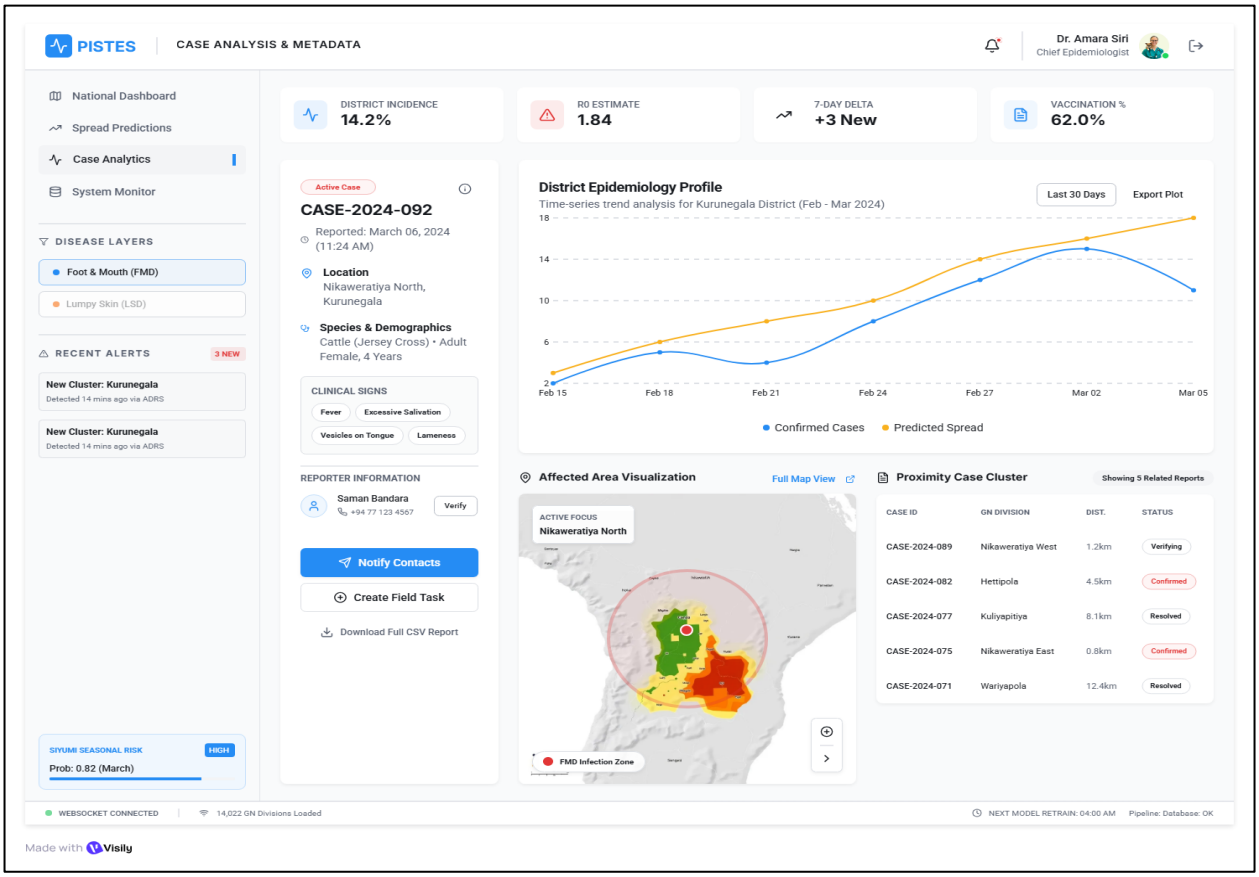


Figure 5 Case Analytics UI mockup

6. USER REQUIREMENTS

6.1 Stakeholder Identification

The proposed ADRS system involves several stakeholder groups who interact with the platform directly or indirectly. The primary stakeholders are **DAPH veterinary officers and district epidemiologists**, who use the real-time dashboard map to monitor outbreaks, identify high-risk areas, and make quarantine and response decisions. The secondary stakeholders are **livestock farmers**, who submit disease reports through the mobile application by uploading animal photographs and selecting farm locations. Additional stakeholders include the **ADRS development team**, which maintains the integrated components, and the **Department of Animal Production and Health (DAPH)**, which serves as the main institutional authority responsible for disease surveillance, regulation, and outbreak response.

Main Stakeholders

- DAPH veterinary officers
- District epidemiologists
- Livestock farmers
- ADRS development team
- Department of Animal Production and Health (DAPH)

6.2 Functional Requirements

FR-01: The system shall allow farmers or veterinary officers to submit livestock disease reports with farm location and animal photograph information.

FR-02: The system shall store confirmed disease case records in a centralized spatial database.

FR-03: The system shall process submitted animal photographs through the Computer Vision component to generate a disease classification and confidence score.

FR-04: The system shall allow veterinary officers to review and confirm disease cases before they enter the surveillance pipeline.

FR-05: The system shall detect active outbreak clusters using ST-DBSCAN based on disease-specific spatial and temporal parameters.

FR-06: The system shall calculate environmental spread direction for confirmed outbreak clusters using wind, terrain, river flow, and livestock density data.

FR-07: The system shall predict outbreak trajectories for Day+7, Day+14, and Day+21 using trained Ridge Regression models.

FR-08: The system shall generate outbreak risk zones using Kernel Density Estimation and classify them into RED, ORANGE, GREEN, and CLEAR zones.

FR-09: The system shall integrate seasonal outbreak probability and severity values from the forecasting component to enhance spread prediction and risk estimation.

FR-10: The system shall display confirmed outbreak clusters, spread trajectories, and GN Division risk zones on an interactive geospatial dashboard.

FR-11: The system shall update the dashboard automatically after each confirmed case without requiring manual page refresh.

FR-12: The system shall generate alerts for newly identified RED-zone outbreak areas for veterinary response.

6.3 Non-Functional Requirements

Performance

The system shall ensure efficient processing of disease reports and outbreak analysis.

• The system shall process confirmed disease reports and update the visualization dashboard in near real-time.
• The complete outbreak analysis pipeline shall execute within **3 seconds** after case confirmation.
• The system shall support simultaneous processing of multiple case reports without significant delay.
• Efficient spatial computation and optimized database queries shall be used to maintain system performance.

Security

The system shall protect sensitive data related to livestock farms and disease reports.

• The system shall implement **secure authentication and role-based access control** for users.
• Different access levels shall be provided for farmers, veterinary officers, and administrators.
• All communication between system components shall use **encrypted HTTPS connections**.
• Exact farm GPS locations shall not be publicly displayed to protect farmer privacy.
• Sensitive data shall be securely stored in the database with restricted access

Reliability

The system shall maintain stable and continuous operation for disease surveillance.

• The system shall provide consistent availability for outbreak monitoring.
• The system shall continue functioning even if certain external services temporarily fail.
• Backup mechanisms shall be implemented to prevent data loss.
• System monitoring mechanisms shall detect and report failures or errors.

Usability

The system shall provide an intuitive interface for users with different technical backgrounds.

• The system shall provide a **simple mobile interface** for farmers to submit disease reports.
• The system shall present outbreak information through clear visual indicators such as colored risk zones and cluster markers.
• The dashboard interface shall be usable by veterinary officers without requiring advanced GIS knowledge.
• Map visualizations shall allow users to quickly interpret outbreak patterns and risk areas.

Scalability

The system shall support future expansion in data volume and system users.

• The system architecture shall support increasing numbers of disease reports.
• The system shall allow expansion to additional livestock diseases and geographic regions.
• The database and processing modules shall support large spatial datasets.
• Cloud-based deployment options may be used to support scalable infrastructure.

Maintainability

The system shall allow easy updates and modifications as new requirements arise.

• System modules shall be designed using a **modular architecture**.
• Disease-specific parameters shall be configurable without modifying core system code.
• Machine learning models shall be replaceable when new training data becomes available.
• Clear documentation shall support system maintenance and upgrades.

Accuracy

The system shall provide reliable spatial analysis and outbreak prediction results.

• The outbreak trajectory prediction model shall achieve measurable prediction accuracy.
• Cluster detection algorithms shall correctly identify outbreak clusters.
• Risk zone generation shall accurately represent areas with higher outbreak probability.
• The system shall demonstrate improved performance compared with baseline prediction approaches.

7. COMMERCIALIZATION PLAN

7.1 Target Market and Customer Persona

The proposed ADRS platform primarily targets public veterinary surveillance authorities responsible for monitoring livestock diseases and responding to outbreaks. In many developing countries, livestock disease reporting systems still rely on manual reporting methods that lack real-time spatial intelligence. ADRS addresses this gap by providing a centralized digital platform capable of real-time disease reporting, geospatial analysis, and outbreak prediction.

Target Market

The system is primarily intended for organizations responsible for livestock disease surveillance and management.

• Government veterinary departments (e.g., Department of Animal Production and Health – Sri Lanka)
• National livestock development authorities
• Ministries of agriculture and animal health divisions
• Veterinary hospitals and veterinary research institutes
• Large commercial livestock farms and livestock cooperatives

Customer Persona

A veterinary academic as well as a public health specialist, Dr. Thusitha K. Karunarathna (Department of Veterinary Public Health & Pharmacology, University of Peradeniya), is an ideal customer persona for this system.

Veterinary Officer

• Responsibilities: monitoring livestock disease cases, outbreak investigation, quarantine enforcement
• Needs: real-time outbreak information, spatial intelligence maps, and early warning indicators

Livestock Farm Manager

• Responsibilities: monitoring animal health and reporting disease symptoms
• Needs: simple tools for disease reporting and access to outbreak alerts

7.2 Value Proposition

The ADRS platform provides significant value by improving livestock disease monitoring and outbreak response capabilities.

The system offers several key benefits:

• **Real-time disease surveillance** allowing authorities to monitor outbreaks as they occur
• **Early outbreak detection** through spatial clustering and predictive analysis
• **Improved veterinary decision-making** using interactive geospatial intelligence maps
• **Reduced economic losses** by enabling faster containment of disease outbreaks

By transforming manual reporting processes into a digital surveillance platform, ADRS helps veterinary authorities respond to outbreaks more efficiently and reduce the spread of infectious livestock diseases.

7.3 Revenue Model

The proposed system can generate revenue through multiple commercialization approaches suitable for public-sector and agricultural technology markets.

Possible revenue models include:

• **Government Licensing Model**
National veterinary departments purchase an annual license to use the system for disease surveillance.

• **Software-as-a-Service (SaaS)**
The system is deployed as a cloud-based platform where organizations pay a recurring subscription fee.

• **Institutional Partnerships**
International agricultural organizations (such as FAO or development agencies) may fund deployment for livestock disease monitoring programs.

These models allow the system to be deployed at national scale while ensuring long-term sustainability.

7.4 Pricing Strategy

The system can be offered using flexible pricing models suitable for government institutions and agricultural organizations.

Possible pricing strategies include:

• **Institutional Licensing**

Government agencies pay an annual licensing fee to use the system nationwide.

• **Subscription Model**

Agricultural organizations or livestock farms pay a monthly or annual subscription for access to the platform.

• **Tiered Pricing Model**

Different pricing packages based on system functionality and usage scale.

Example tiers:

• Basic package – disease reporting and monitoring
• Advanced package – outbreak prediction and spatial analysis
• Enterprise package – full surveillance and national deployment capabilities

7.5 Competitive Advantage

The ADRS system offers several advantages compared with existing livestock disease reporting approaches.

• **Real-time disease reporting capability** replacing manual reporting methods
• **Integrated geospatial visualization** enabling spatial understanding of outbreaks
• **Predictive outbreak modelling** for forecasting disease spread
• **Centralized surveillance architecture** that consolidates multiple data sources
• **Decision-support tools** that assist veterinary authorities in outbreak management

These capabilities make ADRS more advanced than traditional livestock disease reporting systems that only record historical outbreak data..

Table 1: Competitive Analysis of Livestock Disease Surveillance Systems

Feature / Capability	**Proposed ADRS PISTES Component**	**Reporting Systems**	**Global Surveillance Platforms**	**Generic GIS Mapping Tools**
Data Collection	Real-time GPS case reporting from mobile application	Manual reporting via phone or paper forms	Periodic national-level reports	Manual data entry required
Spatial Resolution	Spatial Resolution	Spatial Resolution	Spatial Resolution	Spatial Resolution
Disease Spread Prediction	Disease Spread Prediction	Disease Spread Prediction	Disease Spread Prediction	Disease Spread Prediction
Risk Zone Generation	Risk Zone Generation	Risk Zone Generation	Risk Zone Generation	Risk Zone Generation
Real-Time Monitoring	Real-Time Monitoring	Real-Time Monitoring	Real-Time Monitoring	Real-Time Monitoring
Decision Support	Decision Support	Decision Support	Decision Support	Decision Support

7.6 Intellectual Property Considerations

The ADRS platform contains several intellectual property components that require protection and proper management.

• The system software and source code can be protected under **software copyright laws**.
• The analytical algorithms and system architecture design represent original research contributions.
• Licensing agreements may be used when providing the system to government agencies or organizations.
• Data usage agreements must be established when using external datasets or sharing disease surveillance data.

Additionally, ownership of the research and software implementation will follow the intellectual property policies of the university and the research team involved in developing the system.

7.7 Cost Estimation

The estimated cost of developing and operating the ADRS component includes both initial development expenses and ongoing operational costs.

Development Costs

- Software development and system integration
- Implementation of spatial analysis and prediction modules
- Development of the visualization dashboard and database infrastructure

Infrastructure Costs

- Cloud server hosting for the backend system
- Geospatial database storage and management
- Storage of environmental datasets and system logs

Operational Costs

- System maintenance and software updates
- Model retraining using new outbreak data
- Technical support and system monitoring

A breakdown of the estimated budget for development, training and first year of operation is shown below in Sri Lanka Rupees (Rs.).

Table 2 Cost Estimation

Component	**Cost**
Cloud Infrastructure	Rs. 6,000 / month
Data Storage	Rs. 2,500 / month
Deployment Cost (Inference API)	Rs. 5,000/month
System Maintenance	Rs. 4,000 / month
Total Estimated Budget	Rs. Approx. Rs. 17,500 / month

8. BUDGET AND JUSTIFICATION

8.1 Justification for Each Expense

Hardware Costs

The proposed system does not require specialized hardware for development or deployment. The PISTES component is implemented using software-based spatial analysis and machine learning algorithms that can operate on standard computing environments. Development and testing activities will be conducted using existing university laboratory computers and standard workstations available to the research team. Additionally, the system relies on farmer smartphones for GPS location collection and image submission, eliminating the need for dedicated GPS devices or field hardware. Therefore, no additional hardware procurement cost is required.

Software Licence Costs

The system is developed using open-source software technologies and libraries, including Python programming frameworks, geospatial analysis libraries, and database systems. These tools provide the necessary capabilities for spatial data processing, machine learning implementation, and web-based visualization without requiring commercial software licenses. The use of open-source platforms significantly reduces development costs and ensures that the system can be maintained and deployed without long-term licensing expenses.

Cloud Services Costs

Cloud infrastructure is required to host the centralized surveillance platform and support real-time processing of disease reports. The cloud environment will store geospatial datasets, disease reports, and environmental data while enabling execution of the spatial analysis pipeline. Cloud services also support the deployment of the web-based dashboard and allow veterinary authorities to access the system remotely. Using cloud infrastructure ensures scalability, reliability, and efficient system performance for real-time disease monitoring.

Human Effort Cost

Human effort is required for system design, development, testing, and evaluation of the proposed surveillance platform. Activities include implementing spatial clustering algorithms, integrating environmental data sources, developing predictive models, and building the visualization dashboard. Additional effort is required for system testing, validation experiments, and documentation of research outcomes. The involvement of researchers and technical contributors ensures that the system is properly developed, validated, and prepared for practical deployment.

9. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) for the ADRS system ensures that the research tasks are distributed evenly among the four team members while maintaining a coordinated development timeline. Each researcher is responsible for a specific research component, allowing the system to be developed in parallel and integrated efficiently. The ADRS project is planned to be completed within a nine-month academic period, covering stages from data collection and model development to system integration and final validation..

9.1 Task Breakdown by member

Component 1: AI-Powered Smart Diagnosis (Computer Vision and Prognostics)

- Member 1: A.L.M Athulathmudali
- Key Tasks:

1. Collect and Annotate Primary and Secondary Data Sets for Local Cattle Breeds (i.e. Cattle Breeds found locally) for multiple concurrent diseases and their associated vitality indicators.
2. Design and Build Input Integrity Gate using YOLOv10-tiny and Out-of-Distribution (OOD) Additional Information.
3. Train Hierarchical Multi-Label Classifier (Swin Transformer) and Severity Quantification Module (Mask R-CNN).
4. Implement the Chain of Thought (CoT) transparent framework and prognostic survival engine.
5. Validate the diagnostic threshold using K-Fold cross-validation to ensure it has an 85% accuracy rate.

Component 2: Real-Time Geospatial Clustering & Trajectory Mapping

- Member 2: G.A Sandaru
- Key Tasks:

1. Collect and preprocess geospatial datasets related to livestock disease outbreaks in Sri Lanka, including historical outbreak records, administrative boundary data, and environmental variables.
2. Develop and implement spatio-temporal clustering algorithms (ST-DBSCAN) to detect real-time disease outbreak clusters based on geographic location and time.
3. Apply environmental spread modelling techniques to estimate potential disease spread patterns using environmental factors such as wind direction, terrain conditions, and livestock density.
4. Design and implement trajectory prediction models to forecast possible outbreak movement and identify areas that may become affected in the near future.
5. Develop an interactive geospatial intelligence map to visualize outbreak clusters, predicted spread trajectories, and risk zones for veterinary authorities.

Component 3: Seasonal Forecasting for Risk assessment

- Member 3: S.S Kumarasinghe
- Key Tasks:

1. Combine historical climate records, monsoon characteristics, and environmental variables related to tropical livestock diseases into a comprehensive data set.

2. Design non-linear machine learning forecast models (LSTM's or Conformal Inference) to enable effective temporal forecasting and the identification of sudden spikes in disease occurrence.
3. Create a local, seasonal "Risk Index" by combining relevant weather variables to designate the level of risk within select geographic areas.
4. Confirm the validity of the forecast model's precision when correlating the forecasted outbreak occurrences with recorded outbreak data.
5. Integrate the results of the predictive risk assessment with the central ADRS database to notify vulnerable farming areas.

Component 4: Animal Health Detection (Wellness and Management)

- Member 4: W.M.P.J Wijenayake
- Key Tasks:

1. Develop and establish corporate standard Multiple Base Parameters for the recognition of physiological distress and physical wellness in local dairy cattle.
2. Create a 2D vision-based algorithm to automate the reading of Body Condition Scores (BCS) by using skeletal keypoint detection software.
3. Use the clean segmented masks of bovine animals created using algorithms from the Computer Vision Component to remove lesions when calculating the BCS.
4. Create a farmer view of the wellness of the animal that includes recording the physiological conditions and energy status of the herd.
5. Conduct field tests to compare automated reading of BCS to manually recorded readings prepared by veterinarians.

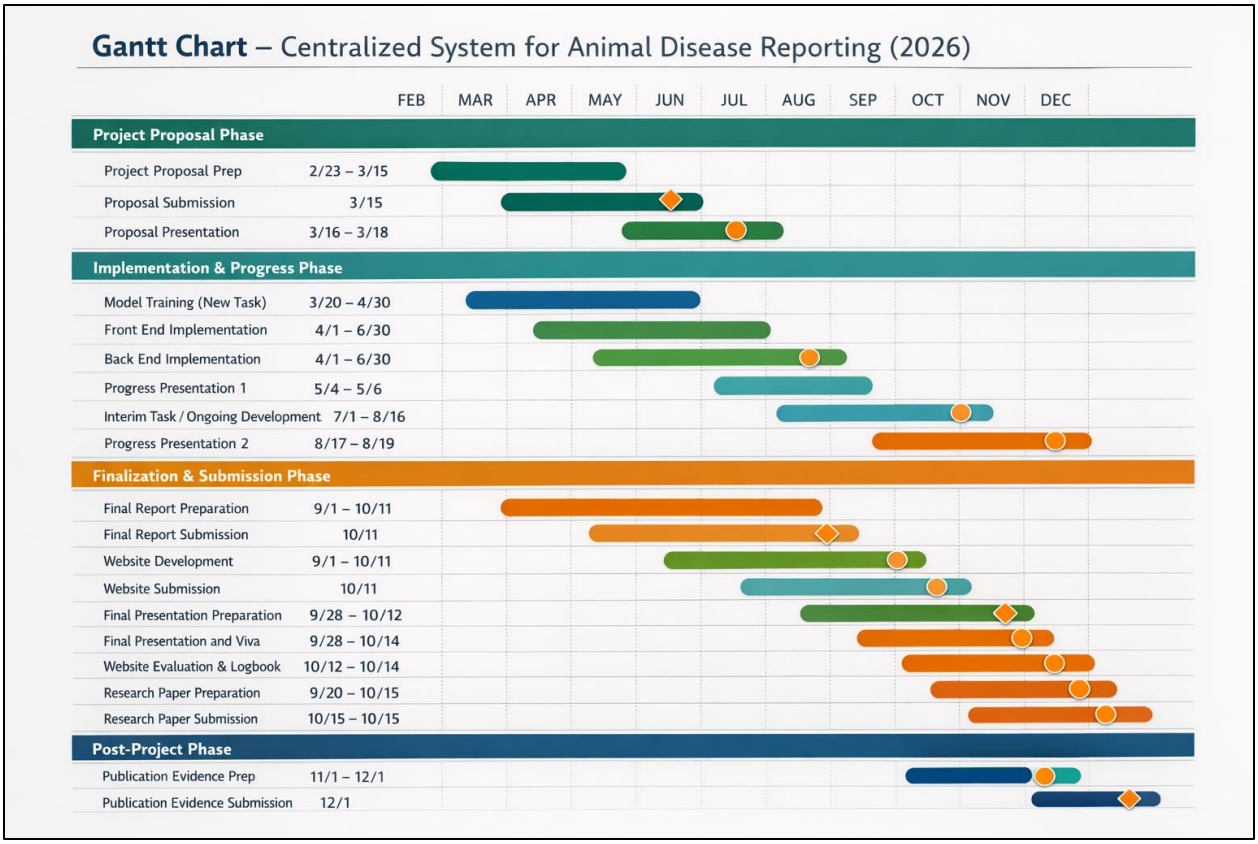
9.2 Timeline and Workload Distribution

To ensure balanced workload distribution, all four research components will be developed concurrently following the same project phases. This parallel development approach reduces

integration delays and ensures that all modules become available at the required stages of system development.

Table 3 Timeline Alignment and Phase Distribution

Project Phase	**Timeline**	**Primary Activities (All Members)**	**Milestones / Deliverables**
Phase 1: Requirements & Data Collection	Months 1 - 2	Literature review, system architecture design, dataset collection, and requirement analysis	Final proposal and initial datasets prepared
Phase 2: Algorithm Development	Phase 2: Algorithm Development	Phase 2: Algorithm Development	Phase 2: Algorithm Development
Phase 3: System Integration	Phase 3: System Integration	Phase 3: System Integration	Phase 3: System Integration
Phase 4: System Testing & Validation	Phase 4: System Testing & Validation	Phase 4: System Testing & Validation	Phase 4: System Testing & Validation
Phase 5: Documentation & Finalization	Phase 5: Documentation & Finalization	Phase 5: Documentation & Finalization	Phase 5: Documentation & Finalization



10.GRANTT CHART

Figure 6 Grantt Chart

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